MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON, TYPE 8405

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1

DESCRIPTION: An electrostatically focused pulsed amplifier Klystron for operation at 5385 to 5900 MHz.

ABSOLUTE - MAXIMUM RATINGS:

Parameters:	Ef	If (Surge)	eb	ib	pi	Pi	pi(rf)	Pi(rf)	Oper. Alt.	Beam	tk
Units:	v	A	kv	a	kw	W	kw	W	Ft.	Du	sec.
Maximum:	6.9	8.0	20	9.4	190	300	0.8	10	10,000	0.00165	--
Minimum:	5.7	--	--	--	--	--	--	--	--	--	120
Notes:	1,2	--	2	--	--	--	17	17	--	2,6(d), 11	1
TEST CONDITIONS:	6.3	--	18	--	--	--	--	--	--	--	--

ABSOLUTE - MAXIMUM RATINGS: (Note 1)

Parameters:	VSWR In	VSWR Out	ib pulse tp	ib pulse tr	ib pulse tf	TA	T Cath. Seal	T Body	T Coll.	Tuner Stop Torque
Units:	Conn.	Load	usec	usec	usec	°C	°C	°C	°C	in-lbs
Maximum:	--	--	3.3	--	--	85	125	100	150	6
Minimum:	--	--	0.1	--	--	-54	--	--	--	--
Notes:	--	--	6a,b	--	--	--	--	5	5	--
TEST CONDITIONS:	2.1 max.	1.3:1	3.0 ±.3	0.6 ±.2	0.6 ±.3	--	Figure 1 Zone A_1	Figure 1 Zone B_1	Figure 1 Zone C_1	-

PHYSICAL CHARACTERISTICS: See figure 1

MOUNTING POSITION: Any

CONNECTIONS: Figure 1

VOLTAGE REFERENCE: Cathode

GROUND: Tube Shell

COOLING: Forced Air; Note 5

TUNING: Note 4; Figure 1; and Tuning Shaft Characteristics (listed under Quality conformance inspection, part 1)

DIMENSIONS: Figure 1

START-UP PROCEDURE: Note 2

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection (Note 10)					
---	Storage	No Voltages	---	---	---	---
---	Storage end points	Po(1) Note 7				
4251	Power Output for all load phases	Test condition; Vary load through all phases Notes 5, 6, and 9a.				
		Load VSWR				
		1.3:1	po	25	--	kw
		1.5:1	po	18	--	kw
		Infinite Load Mismatch		No damage		
---	Temperature	Test condition; pi (rf) = 200W Temp. range 30 to 85° C TA; Note 15 F = 5385 and 5900 MHz	Δpo	--	4.0	dB
4223	Mechanical tuning fatigue	No Voltages; Note 12		300	--	cycles
	Mechanical tuning fatigue end points	tuning shaft characteristics				
1042	Shock	No Voltages; Condition H; Note 13				
---	Vibration	No Voltages; Note 14				
---	Shock and vibration end points	Beam current & Emission				
	Quality conformance inspection, part 1					
1231	Emission	Test Cond.; pi (rf) = 0 Note 3 and 21	Δib / ib	--	10	%
	Tuning Shaft Characteristics (a) Turns ove · freq. range	Test condition; pi (rf) = 200W Note 16	po	25	---	kw
	Input and Output Cavities			9	13	turns
	Middle Cavity			8	12	turns
	(b) Turns Between stops	No Voltages				
	Input and Output Cavities			--	18	turns
	Middle Cavity			--	19	turns
	(c) Tuner Shaft Torque Turning	No Voltages		--	5	in-lb
	Against stops	Apply 6 in-lb.		No damage		
1301	Heater Current	Ef = 6.3V; eb = 0	If	1.9	2.3	A
1256	Beam Current	Test Cond.; pi (rf) = 0; Note 21	ib	--	8.5	a

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 - continued					
4250	Power Output (1)	Test Cond.; pi (rf) = 200W F = 5385, 5650, 5900 MHz Notes 5, 6, 9(a), 4	po(1)	25	--	kw
4253	Low Level Gain	Test Cond.; pi (rf) = 50W max Note 8	Gss	24	--	dB
	Quality conformance inspection, part 2					
4256	Input Coupling	Test Cond.; pi (rf) = 200W Note 18	VSWR	--	2:1	
4271	Bandwidth	Test Cond.; pi (rf) = 200W; Notes 6 and 9		25	--	MHz
4266	Direct Interelectrode Capacitance	Electron tube gun to tube body	Cak	6	12	pF
	Quality conformance inspection, part 3					
---	Life Test	Test Cond.; pi (rf) = 200W F = 5800 ±25 MHz Group D; Notes 19 and 20	t	1000	---	cycles
---	Life Test End Point	Power Output (1)	po	---	20	kw

NOTES:

1. Some absolute ratings are incompatible. Attempts to operate at such ratings simultaneously may damage the tube.

2. New tubes, or tubes stored over 3 months require a special start-up procedure as follows:

   (a) Apply heater voltage.
   (b) After heater warm-up, apply approximately eb = 6 kv for 30 minutes, then apply approximately eb = 12 kv for 15 minutes, after which operational voltages may be applied.
   (c) Tube start-up procedure for normal operation: Apply Ef for time tk.

3. Under conditions in beam current test (method 1256) the heater voltage is decreased from 6.3 to 5.7V. The percentage change in cathode current, after 1 minute shall be not more than the value specified.

$$\frac{ib\ (Ef = 6.3\ V - Ef = 5.7\ V) \times 100}{ib\ (Ef = 6.3\ V)}$$

4. (a) A plasticized tuning curve shall be supplied with each tube. The frequency of each resonator shall be set by adjusting the appropriate tuning screws according to the tuning curve. The frequency of each resonator shall be within ±25 MHz of the frequency specified in the tuning chart.
   (b) Clockwise rotation of the tuning screws shall increase the frequency of each resonator.

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5: Sufficient cooling shall be distributed evenly over the entire tube so that maximum temperature ratings are not exceeded. Recommended cooling at sea level, when operated at maximum power input (Pi), is 100 C.F.M. The maximum temperature found in each zone shown on figure 1 shall apply.

6: (a) Beam pulse width shall be measured between the 50 percent points of the beam current pulse when viewed by a linear circuit which presents a voltage representation of the pulse. For illustration of the pulse see MIL-STD-1311, method 4304, figure 4304-1.

(b) The beam current pulse characteristics (when viewed as specified in Note 6(a)) shall be measured at the pulse amplitude points indicated below:

Pulse Top:	90 percent
Rise Time:	10 - 90 percent
Decay Time:	90 - 10 percent

(c) R.F. pulse width shall be measured between the 3 dB points of the r.f. output pulse when viewed through a crystal detector operating in the linear region or an equivalent circuit.

(d) The maximum operational P.R.F. shall be determined by the maximum duty cycle and the pulse width applied.

7: Storage - The tube shall be capable of storage in shipping containers at air temperatures within shipping containers from -54° to +71°C without impairment of subsequent performance. The tube, furthermore, shall be capable of meeting all requirements specified herein after 50 minutes of initial warm up. The above shall be demonstrated by means of the following tests:

(a) Cold cycle - with the tube in the test chamber and beginning at laboratory ambient temperature, the temperature of the chamber shall be cycled from +21° to -54°C. and back to +21°C. This cycle shall be repeated five times. The -54°C temperature shall be maintained for at least 3 hours during each cycle. Also, the +21°C. temperature shall be maintained for at least 3 hours between cycles. After the completion of the required cycles the tube shall be stabilized at the ambient laboratory temperature and then tested for satisfactory performance.

(b) Hot cycle - Starting at laboratory ambient temperature, the temperature of the chamber shall be cycled from +21 to +71°C and back to +21°C. This cycle shall be repeated five times. The +71°C temperature shall be maintained for at least 3 hours between cycles. At the conclusion of the last cycle the tube shall be stabilized at ambient laboratory temperature and then checked for satisfactory performance.

(c) Either the hot or cold cycle may be performed first. The rate of temperature change of the test chamber from +21° to either -54° or +71°C shall be accomplished in no more than 3 hours. Likewise, the change from +71° or -54° to +21°C. shall require no more than 3 hours.

8: (a) When the three resonators of the tube are identically tuned to the r.f. drive frequency the tube is synchronously tuned.

(b) The tube shall have all resonators tuned identically to the drive frequency at the specified eb.

(c) The value of peak r.f. output power obtained under synchronously tuned conditions shall be used to determine the low level gain.

9: (a) Tune the input and output resonator to the drive frequency specified herein and detune the middle resonator toward frequencies greater than the specified r.f. drive frequency until the power output is optimized for the specified r.f. drive power.

(b) The bandwidth shall be measured at the 3 dB power output points under power output test conditions.

10: For qualification testing all environmental tests shall be performed prior to electrical tests.

11: For a maximum eb of 15 kv a Du of 0.004 is permitted.

12: The tube manufacturer may use tubes in the performance of this test that do not meet all of the requirements contained herein, excepting those tubes that do not meet the specified frequency range.

13: Shock - The tube shall withstand the application of two blows in each of three mutually perpendicular planes or equivalent when mounted in a manner that simulates the mounting shown on Figure 2 such that the tube and mounting fixture have a natural frequency of 31 ± 3 Hz with a maximum deflection of 3/8 inch.

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14: Vibration - The tube shall withstand the application of type I vibration in accordance with MIL-STD-167 except that nonresilient mountings shall be used and the amplitudes from 22 to 33 Hz shall be three times those specified. The tube shall be mounted in the fixture shown in figure 2.

15: With the tube in the environment chamber, adjust to obtain the test condition. Turn off the tube and reduce the chamber temperature to the specified low value and allow to stabilize for two hours. Start the tube and measure the power output. Turn off the tube and increase the temperature to the specified high value. Start the tube and measure the power output. The difference in power level shall not be greater than the specified maximum limit.

16: Tune the tube to 5385 MHz and measure the power output. Remove the beam voltage and retune the tube approximately one turn of the tuning shafts higher in frequency. Measure the power output. Repeat at successively higher frequencies until the frequency 5900 MHz is reached. The power output shall be not less than the value specified and the number of shaft turns shall be within the limits specified.

17: pi (rf) and Pi (rf) are the incident powers to the input r-f terminal of the tube.

18: With the tube tuned to 5385 MHz, apply voltages to obtain the test condition. Measure VSWR and turn off all voltages except heater voltage. Repeat procedure at 5500, 5600, 5700, 5800 and 5900 MHz. VSWR at each frequency shall be not more than the value specified.

19: A cycle shall consist of the following events:

Operation under Test condition for 30 minutes
Standby with heater power only for 15 minutes

20: Tubes selected for this test shall have passed Quality Conformance Inspection, Part 1.

21: This test to be performed at the conclusion of the holding period.

User activities:
Navy - AS, OS, MC, CG, SH,

Preparing activity:
Navy - EC

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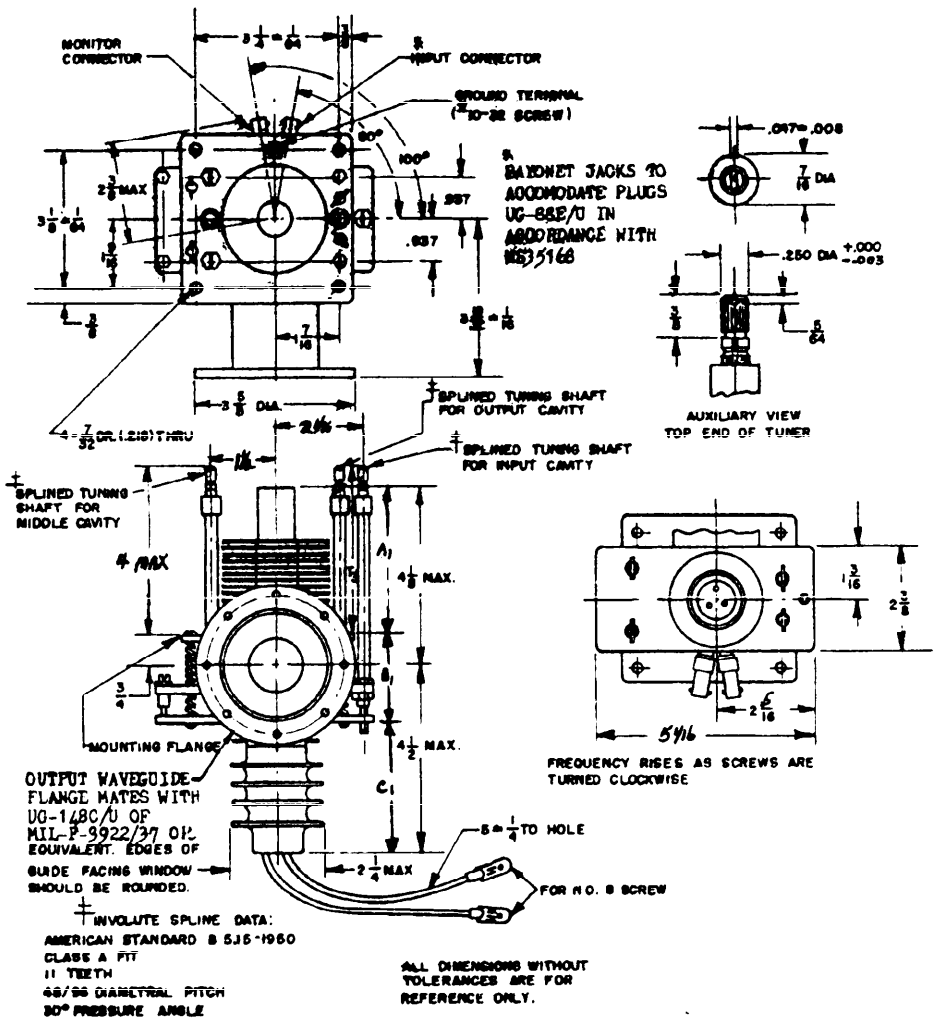


FIGURE 1. Outline drawing.

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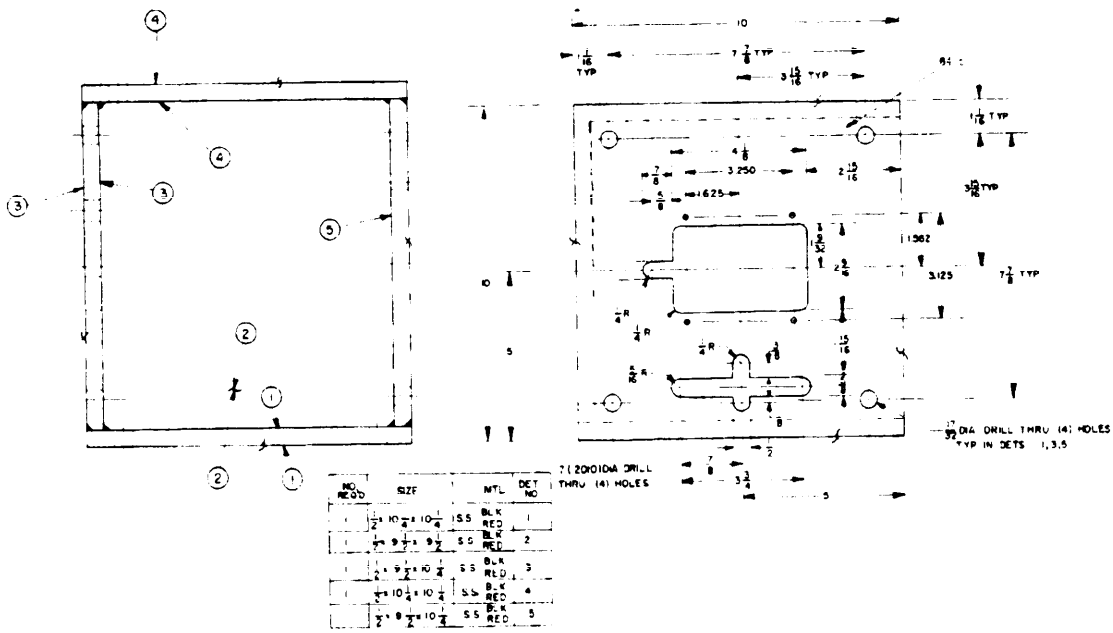


FIGURE 2. Outline drawing.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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